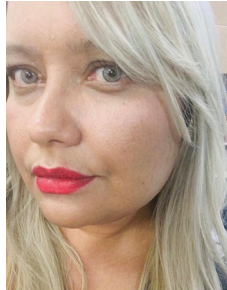


Digital twin development for airport management

Received (in revised form): 15th February, 2020



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Abstract

The Digital Airport and Building Information Modelling (BIM) technology linked to planning, project and management processes, construction, maintenance and operations in parametric three-dimensional design with all the underground, land and above-land information serves as a parametric database for supporting planning and improving project management. An airport digital twin is not limited to building infrastructure; it includes the associated working processes. It serves as a large-scale information repository to improve performance and obtain data more efficiently. All business sectors will benefit, including the commercial, assets management, operations, engineering and maintenance departments. The model will also promote an environment that will lead to digital transformation. For the implementation of such digital models effectively in the company's business, strategic planning is necessary so that digital twins can be used well by organisations to bring good results to the company.

Keywords

digital twin, building information modelling, strategic planning, IoT, airport digital transformation

INTRODUCTION

A digital twin is a realistic digital representation of something physical. A digital twin must represent physical reality at a level of accuracy suited to its purpose. It is an integrated model of an as-built product, including the physics, fatigue, life cycle, sensor information, performance simulations etc. It is intended to reflect all manufacturing defects and be continually updated to include wear and tear sustained while in use.¹

Research has been done on digital twins in a more theoretical way. The

digital twin is a virtual representation that interacts with the physical object throughout its life cycle and provides intelligence for evolution, optimisation, predictions, analyses etc.

According to Fei Tao,² four parts can be considered as the core of the digital twin, namely model, data, connections and services. The model reproduces the real properties, behaviours and rules of the physical equivalent with high fidelity and thus is an exact digital duplicate that records all changes of the physical one. Data can be considered as a driver

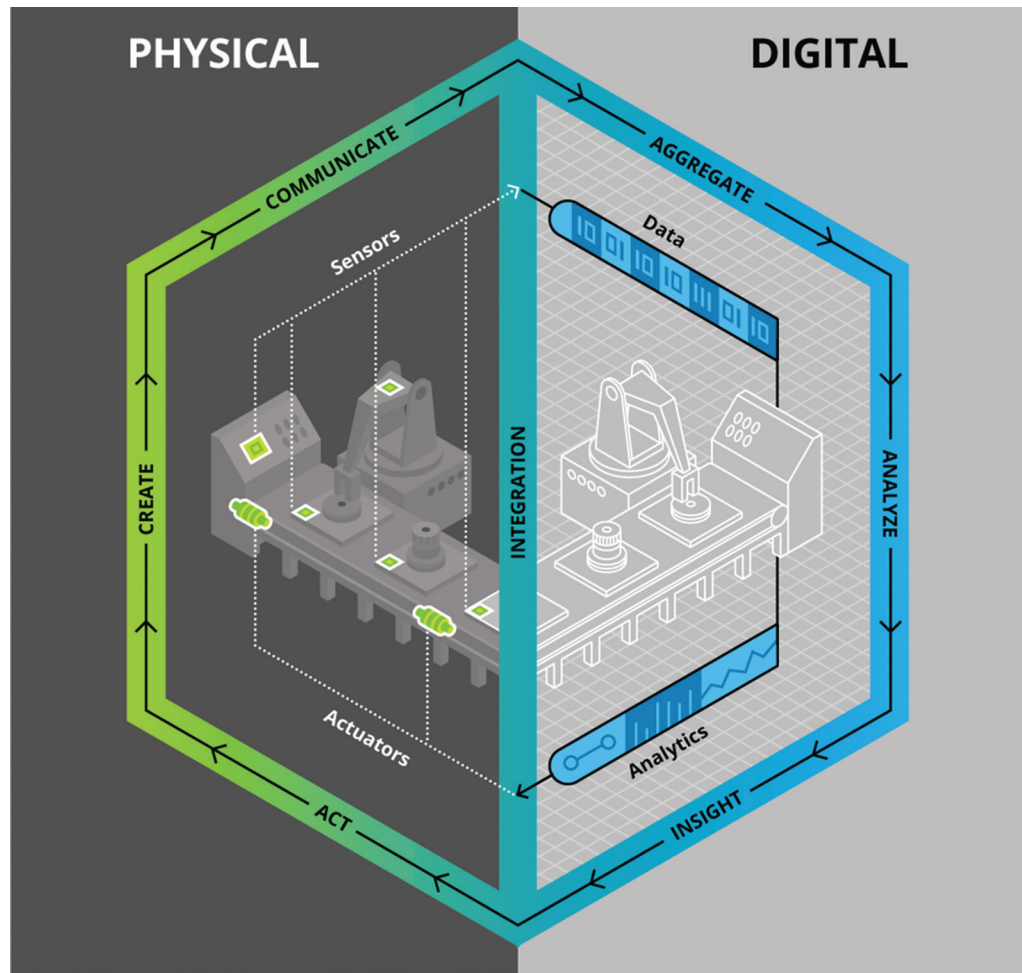


Figure 1 Manufacturing process digital twin model
Source: Deloitte University Press.

that provides intelligence to the digital twin. The connections enable every element in the digital twin interact with each other. The services are the final presentations of the digital twin, which are generally expressed in standard formats to describe their inputs, outputs, basic information and functions, according to the demand.

Figure 1 shows the interaction of the physical environment with the digital environment sensors to make continuous measurements that are transmitted to a digital platform, which performs analyses

in almost real time to optimise a business process transparently.

DIGITAL TWINS: ECONOMIC BENEFITS

Digital twins link physical systems and virtual spaces on the internet of things. They connect physical systems to the digital world while collecting data from an asset to create business value.

The value of the digital twin can be seen by looking at the entire life cycle of the asset. For this, a strategic definition

of the asset will help to reduce business risks, data management can improve collaboration and the digital twin can be used to perform predictive analysis and help reduce unscheduled downtime. Prescriptive maintenance will improve safety and reliability in maintenance and lower costs.

The value of the digital twin also helps making decisions based on more reliable information. Organisations based on a culture of silos can suffer from breaking down of silos. The lack of collaboration between teams is expensive. Asset data management can promote making more accurate decisions.

Using the digital twin can help break down information silos within an organisation and help obtain the right information at the right time. Inspection data, work history and extremely large datasets can be used in the context of a digital twin. Using extensive graphics and dashboards and real-time and reliable information helps to reduce the total cost of assets; better manage, make changes and obtain accurate returns; and ensure better asset performance and return on investment (ROI).

Asset managers will have to add software skills to their operations teams as they add smarter assets and address more complex digital twins in their operations. They must also add software terms and data to their contracts.

As such, the costs of developing and maintaining digital twins must be driven by economic and commercial models. The concept and business model must be tested against an economic model — revenue, profits, ROI, cost optimisation — with defined performance indicators to measure progress as products/services are being delivered and implemented.

The different levels of complexity of digital twins — from simple devices like light sensors to complex assets — will have different ongoing development and maintenance costs, sophisticated chief information officer (CIO) design business cases and ROI analysis.

Issues such as digital ethics and property rights raised by different parties that interact with data not only from the company but also from its partners and customers should be considered. In some cases, complex systems require multiple digital twins with different business requirements to meet specific needs.

Complex assets are usually composed of several digital twins, usually organised into large composite digital twins. This will drive deployment development, integration and cost analysis because at each layer of a composite digital twin, there will be different business requirements to meet the needs of different constituents (eg manufacturer, customer and maintenance provider) that will be added to and increase the complexity, behaviour and cost of the digital twin.

AIRPORT DIGITAL TWINS' PURPOSES

Airport digital twins can be developed for different purposes, operate at different scales or take different approaches to modelling. Several airport digital twins have begun to appear in the built environment, serving various purposes. Few digital twins are, however, currently connected or sharing data between organisations or geographic data. Lack of interoperability is an important constraint.

The reasons to use digital airports for airports are:

- To create a data-based model of assets or processes, paving the way towards

advanced techniques to anticipate and prevent equipment failures and find ways to improve performance. To capture data from old technology still in use and discover tendencies and patterns from it that feed into digital twin models.

- To obtain detailed information about large groups of key assets in complex systems so that it can help determine the best way to allocate scarce investment. This also highlights gaps and allows for informed and differentiated decisions to be made about where to invest in new assets or upgrade existing assets.
- To establish a culture of data-based decision-making, consequently supporting all sorts of ways to improve performance. Also, it helps perform more predictive and condition-based maintenance, which help to improve performance in the long term. Anomaly detection in place in digital twins can detect problems far earlier. This permits planning rather than reacting when problems arise. The proactive approach also allows businesses to prevent problems from occurring in the first place. Using digital twins provides a better way to manage airport business with more insights and data-based analysis.

DIGITAL TRANSFORMATION

The use of digital twins must be tied to the airport business that inevitably must go through a process of digital transformation. This transformation often requires profound changes in the business. Organisational culture needs to be reviewed and is one of the main factors influencing the transformation process.

As technology changes very rapidly, organisational change and the way

it thinks and behaves are still slow and difficult. Many companies think they are transforming but do not understand that transformation must be a permanent state.

Often, leaders find it difficult to change, who are afraid of new technologies and possess the reluctance to innovate and perform the digital transformation, which requires a whole new way of doing business. Digital transformation must be a continuous process of change in the way an organisation does business. It involves investments in people, infrastructure and information technology systems. The company must be concerned with monitoring change to ensure that the digital transformation is meeting its strategic goal.

Many organisations are looking to transform the way they operate, according to a survey of 734 leading companies conducted by Harvard Business Review Analytic Services in July 2018.³ It has been shown that only 13 per cent of respondents rate their digital transformation efforts to date as effective.

It has been realised that although it is possible to progress at the technological and procedural levels, there is still difficulty in changing the organisational culture.

WHAT IS DIGITAL TRANSFORMATION? AN APPROACH TO AIRPORTS

‘Digital transformation is about business transformation in a digital world.’

Digital transformation is not only about technology. It is also about business transformation in a digital world. It involves both the implementation of new technologies and the integration of existing technologies, processes and services.

For the development of a digital airport, it is first necessary to understand the

context in which the airport is inserted. The basic steps for developing a digital airport are:⁴

- Airport environment assessment: It consists in understanding the characteristics of the airport, market, local community and socioeconomic influences.
- Airport strategic plans and objectives: A multidimensional review of the airport's characteristics, including but not limited to its physical, customer, market, local community and economic characteristics and socio-political influences upon it.
- Internal organisational reviews and requirements: A specific step for airport management to agree on the priority and actions of specific digital plans and identify areas where technology (existing or new) will improve the outcome or desired results.

The focus of digital transformation at airports involves the deployment of new technologies as they evolve and integrate them into existing systems and processes. It is a search for a better experience for all stakeholders involved in the airport business.

Transformation aims to integrate systems and services provided by partners such as airlines, customs, security, commercial, automated systems etc. An example of the impact of digital transformation is the use of technologies for geolocation, passenger flow, wayfinding, utility management, data capture, data mining, beacon usage and Internet of Things (IoT).

Many airports around the world are already making some transformation in their processes to maximise their business and operational objectives. A range of data is processed in the airport system, such as generating a lake of data that can be transformed into information,

generating important histories and statistics to verify trends and making them a great support for airport business decision-making.

Artificial interconnection can be used to enhance the passenger experience through algorithms targeting marketing campaigns based on the airport's passenger geolocation, time remaining before boarding and social networks. A digital transformation that really drives changes in the organisation is not only focused on the use of technology with digital mobile applications or remodelled websites. A digital transformation is capable of enabling technologies such as through:

- Having a digital infrastructure ready to build all digital resources
- Embracing the concept of open data and share data where it adds value
- Offering passengers custom experience on their journey
- Using biometry-enabled digital touch-points to make the passenger journey more perfect
- Gathering all relevant data together for all stakeholders in one virtual control room and generate data through IoT
- Actively looking for innovation to generate even more value for customers and monetise technology solutions in new business models

Operationally, digital transformation involves automating and monitoring critical airport systems such as:

- Lighting
- Air conditioning
- Hydro sanitary
- Luggage track
- Elevator
- Escalator
- Sewage

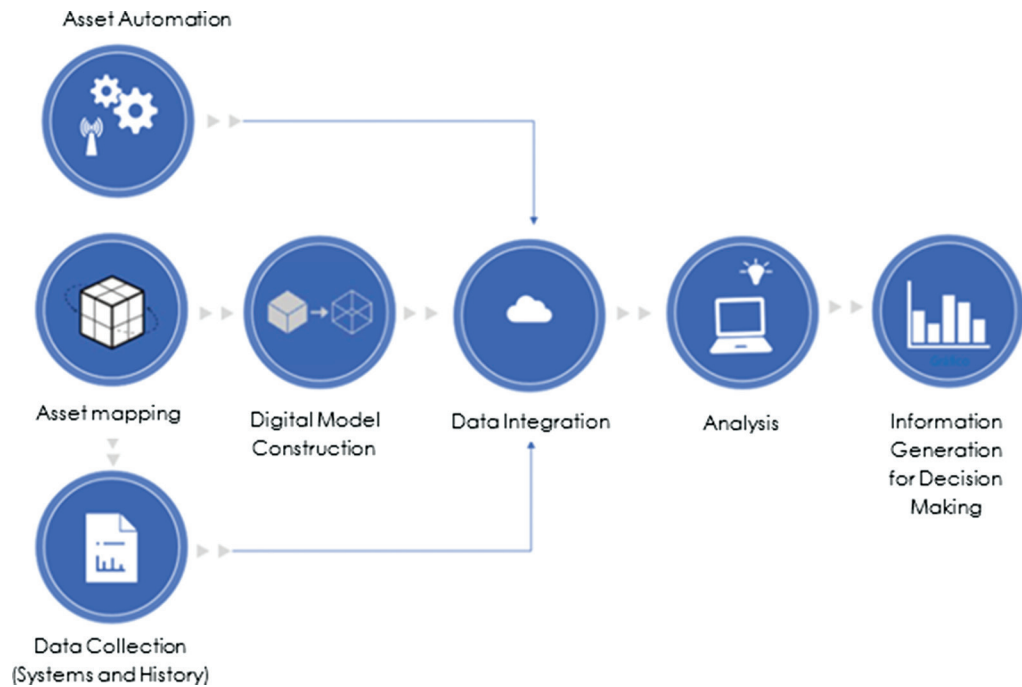


Figure 2 Digital asset transformation

- Access control
- Substation

The amount of physical assets available at the airport, and their importance for the continuation of airport business, has sparked interest in IoT-based solutions, which connect physical systems to the digital world while collecting data from an asset to create business value. The value is increased productivity and efficiency in line with the airport's business model.

In the digital twin system, there should be data integration between automation and data collection, allowing analysis and thus information generation for decision-making (see Figure 2).

CREATING A DIGITAL AIRPORT

A digital airport is the union of a digital twin, aligned with organisational processes and use of new technologies for the

benefit of all stakeholders. In the context of digital transformation, the development of a digital airport is based on three main pillars: data science, connectivity and virtual model, which are represented in Figure 3.

Virtual model: Building information modelling

The National BIM Standard (NBIMS)⁵ defines building information modelling (BIM) as an acronym that is used to describe three distinct, but linked functions:

- Building information modelling:** It is a business process for the generation and delivery of the project and construction and operation of a building during its life cycle. BIM allows all stakeholders to have the same information at the same time through an interoperable platform.

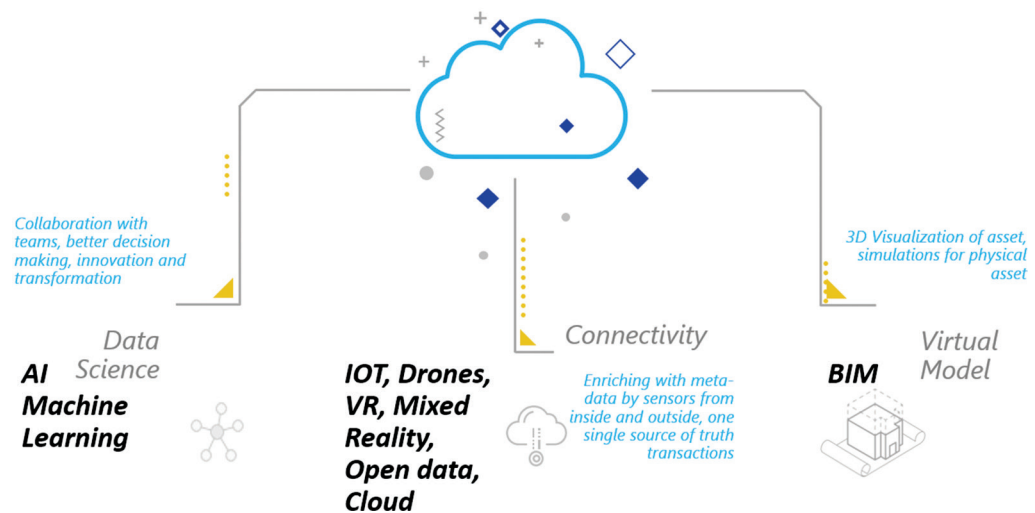


Figure 3 Pillars for the development of a digital airport

Note: AI, artificial intelligence; BIM, building information modelling; IoT, internet of things; VR, virtual reality.

- b) Building information model: It is the digital representation of the physical and functional characteristics of an installation. It is intended to serve as a shared information resource for obtaining data about a facility, providing a reliable source for decision-making throughout the asset's life cycle.
- c) Building information management: This is the organisational and control of the business process, using information from a digital prototype and sharing it throughout an asset's life cycle. It combines the first two functions for organising and controlling the building and infrastructure. Benefits include centralised and visual communication, initial exploration of options, sustainability, efficient design, integration across multiple areas of the company and site control as built documentation.

The first step in adopting and deploying BIM in an organisation, like an airport, is to clearly identify the uses and objectives of implementing the methodology. The next step is to determine which

BIM processes and tools will enable these goals to be achieved. According to Kreider and Messner (2013),⁶ an organisation can use BIM to gather, generate, analyse, communicate and consolidate information about an asset. The subsequent list outlines these goals and their subcategories:

Gather: To collect or organise information about an asset:

- Capture the current status of an installation and parts of it.
- Quantify the value of an installation.
- Monitor the performance of an installation and the systems involved.
- Qualify the status of the installation elements.

Generate: To create installation information:

- Prescribe the need and select specific elements of the installation.
- Determine the location and placement of installation elements.
- Scale the magnitude and scale of the installation elements.

Analyse: To examine the elements of an installation and gain a better understanding of it:

- Coordinate the efficiency and harmony of the relationship of the elements of the installation.
- Predict the future performance of the installation and installation elements.
- Verify and validate the accuracy of the installation information if it is logical and reasonable information.

Report: To display information about an installation and the method by which it can be shared:

- View a realistic representation of an installation or elements of it.
- Transform information and translate it to be received by another process.
- Draw a symbolic representation of the installation and its elements.
- Document the installation information, including the information necessary to accurately specify its elements.

Execute: To create or control a physical element using installation information:

- Fabricate the elements of an installation.
- Assemble the separate elements of an installation.
- Control the operation of execution/handling equipment.
- Regulate the operation of one element of the installation.

The development of the BIM virtual model requires the preparation of an organisational implementation strategy. A BIM deployment committee should be set up to develop strategic deployment planning and advance the actions

defined in the strategic plan. Organisational assessment of the stage in which it is in relation to the BIM maturity level is very important so as to check the areas that need the most attention and consequently define the actions to advance the implementation. The deployment committee must decide the desired level of BIM deployment to establish BIM goals and objectives, BIM uses and established desired levels of implementation maturity.

With the organisational evaluation and the establishment of the planning elements, it is necessary to develop an advancement strategy, which consists in monitoring the implementation, mitigating risks, controlling costs and measuring progress and effectiveness of actions. A baseline should be established to track progress over time-bound milestones. It is necessary to develop a BIM implementation team with defined competencies and responsibilities for the execution of BIM.

BIM's main use for the digital airport is the 3D visualisation of airport critical assets, simulations and basis for airport asset management:

1. Approach and takeoff analysis: detection of plant interference and area analysis such as cities and new constructions, controlled view of the airspace corridor, solar analysis and radar shadows.
2. Airport data management: inspection/condition management, brands, thematic display, NAVAID, catalogue data, BIM data.
3. Predictive analyses: systems of feeding and lighting, chiller/mechanical systems, escalators, hiking, elevators.
4. Information management: single source of information, secure access to data, application integration, document

encoding, document attributes, workflow management, geospatial data management, web accessible and mobile devices.

5. Common data environment (CDE) is a single source of information used to collect, manage and disseminate documentation, the graphical model and non-graphical data to the entire project team (ie all project information created in a BIM environment or conventional data format).

Creating this single source of information facilitates collaboration between project team members and helps to avoid duplication and errors.

Benefits include:

- Connected data environment, enabling real-time collaboration
- Single repository of information
- Improvement of internal project management, including remote works
- Definition of performance indicators
- Control of documents and standards
- Construction of panels for monitoring asset management

Connectivity

The use of drones, virtual reality, augmented reality and mixed reality brings together real and virtual elements, where it is possible to interact with and manipulate items and physical and virtual environments, using detection and image technologies, enriched with meta-data obtained by sensors from inside and outside, a single of truth transactions.

To transmit physical asset data to the end device, various components of IoT technology need be connected, adapted and integrated into the existing information technology (IT) and operational technology (OT) landscape.

Key components of IoT technology include sensors, devices, communication networks, cloud infrastructure, IoT platforms and applications.

Data science

The complete integration of data from automation sensors and critical airport-related asset management systems allows the construction of business intelligence models as well as the development of failure prediction models to reduce costs and impacts of unplanned interventions.

Holistic asset management using a combination of different data sources allows for operational improvement and continuous monitoring of asset availability. The combination is a dynamic and interconnected set of data that features airport asset management capabilities aimed at improving the passenger experience.

CASE STUDY: EXISTING FACILITY AIRPORT DIGITAL TWIN LONDRINA — BRAZIL

The development of the first phase the digital twin at Londrina Airport had as one of its goals the integration between the airport as is, a reliable record of the existing facilities and the geographic information system (GIS). The representation of the real world in a GIS system, together with a database pointing out all the facilities of the airport site aimed to facilitate the analysis and study processes of the airport and provide a link between the various airport systems, allowing a more efficient management of assets and reducing the time and investment costs required.

For the development of the project, three phases were defined, with their

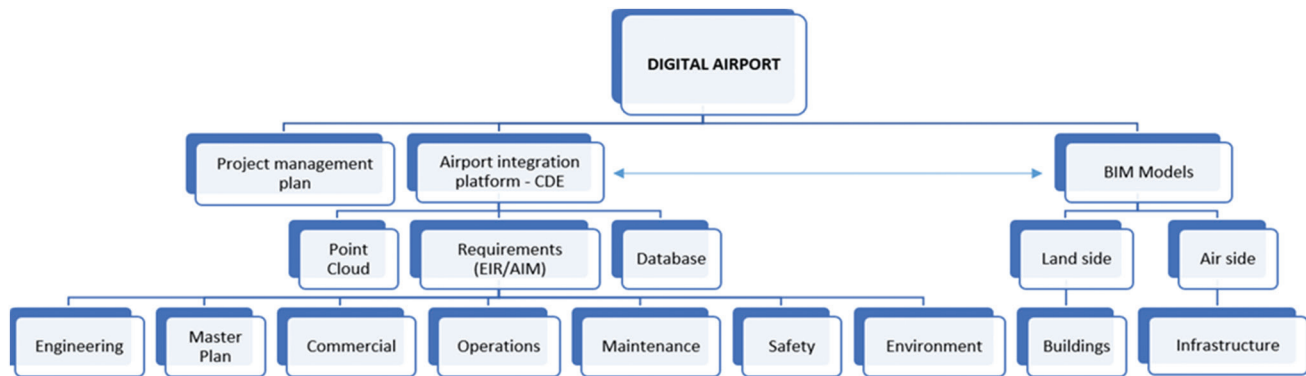


Figure 4 Analytical structure of the digital airport development project

Note: AIM, asset information model; BIM, building information modelling; CDE, common data environment; EIR, employer information requirement

respective deliveries: (1) the project management plan, (2) airport integration platform and (3) 3D modelling of the Londrina airport site. Figure 4 shows the analytical structure of the project.

The digital airport project was developed following the management principles of the Project Management Institute (PMI). A work plan was carried out with the list and description of the activities to be developed; for that purpose, an analytical project structure (WBS) and its dictionary were developed.

For the monitoring of the work, a schedule for the execution of the activities was defined, within a period of 18 months.

The technical project team is composed of about 30 professionals from various areas of the company, distributed in all regions of the country. They are professional specialists in several areas of knowledge, such as architects, engineers (civilians, electricians, electronics, mechanics, cartographers) systems analysts, air navigation specialist, building technicians and designers. These professionals are organised in two groups: integration platform development group and parametric modelling group.

The phase that required the greatest work effort for the greatest integration of the areas involved was the gathering of data on the airport.

Information such as patrimonial areas and limits, validated information regarding the physical characteristics of the aerodrome (runways, taxiways, aircraft mix), air navigation equipment, safety information and airport systems was obtained. There was some difficulty in obtaining the information quickly, in addition to unique information about the same asset. This problem was due to the fact that the different departments work in information silos; there was some initial resistance in making information available.

The team responsible for the project did a convincing job of demonstrating the benefits of obtaining shared data. With single centralised information available to all, there would be a reduction in rework due to missing or conflicting information as well as the improvement of the process of airport management, consequently resulting in better management of the company's business portfolio based on transparency, reliability and traceability and generating greater legal, administrative and management reliability.

Enabling subsidies for the planning of operations (master plans) and creating an environment through the BIM/geographic information system (GIS) interfaces created according to the requirements of the demanding areas.

A common airport data environment was intended to promote the following benefits in each of the areas:

Engineering

- Reliable registration of airport infrastructure
- Standardised design process developed using the BIM methodology
- Monitoring and control of assets in real time
- Greater efficiency of remote asset management
- Management optimisation aiming to balance costs, risks and opportunities

Master plans

- Integrated master plans
- Zoning and land use
- Airport protection plans

Commercial/business

- The dynamic commercial mix (retail and advertising)
- Potential assessment of commercial areas (internal and external)

Operations

- The airport geographic information system
- Dynamic demand and capacity assessment (terminals, patios and runways)
- Simulation and assessment of aircraft movements
- Studies for infrastructure expansion

Security

- Study and monitoring of passenger flows

- Emergency operations optimisation
- Escape routes

Environment

- Monitoring of environmental requirements
- Energy-efficiency assessment
- Water resource management
- Waste management and atmospheric emissions

The airport model consisted of the 3D modelling work of the airport site and its buildings, from cadastral plans and point clouds using BIM modelling software.

Before the modelling work, a survey was made of all drawings available at the airport in Londrina, with the aim of having a register or documents containing the necessary information on the floors, buildings and facilities existing in the place that covers the scope of the project. These registration and data collection actions were carried out to generate the basic and reliable information for the modelling of the digital airport, including connections with utilities for energy and water from information made available through the city of Londrina.

Part of the team was responsible for the parametric modelling of the areas comprising the airport site, presenting information about the dimensions regarding the runway and take-off system and aircraft yards and taxi runways (width, thickness and lengths) and also about the types of pavements; concrete plates for rigid pavements, manholes for sewage networks and rain-water galleries, wolf mouths, gullies and other drainage components; existing reservoirs and the entire distribution network; treatment systems or existing public network, boxes and postage of

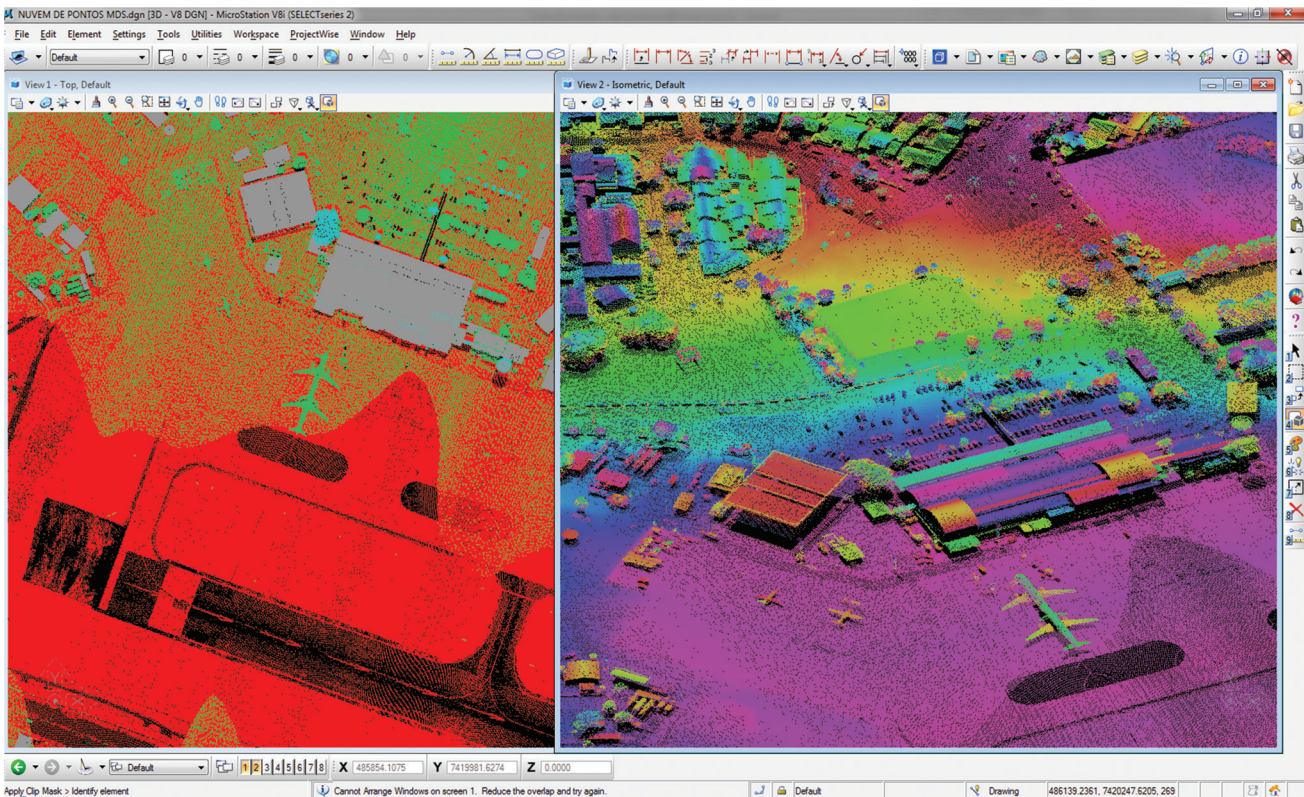


Figure 5 Digital surface model (MDS) from a point cloud obtained by laser scanning

electric network; information regarding the parking of vehicles and accesses; land-side road systems and accesses; location of trees, posts, hydrants and other existing built elements; location of sidewalks and accesses; and also the registration of landscaping with information in the area of landscaping, plant species, and the location of trees and species specification.

A topographic survey was carried out through an aerial survey, providing one of the products in an digital surface model (MDS) file in order to enable the generation of a point cloud with the elevation of the terrain and the location of the existing infrastructure on the site airport of Londrina.

Figure 5 shows the result of importing the .LAS file in Microstation V8.

From this import of the point cloud, it is possible to extract various data on the aerodrome's physical characteristics, such as longitudinal profile and cross-sections of the runway and taxiways, terrain slopes, and carrying out studies of cut, fill and regularisation of prepared strips and analyses for drainage systems, among others.

One 3D modelling was done of the land side of the airport site, containing the necessary parameters for the identification of the entities. Among the buildings that were modelled after the predefined libraries were the passenger terminal and main equipment (escalator, luggage conveyor, elevator).

The first phase of delivery of the digital twin of Londrina, which consisted of delivering a digital model of

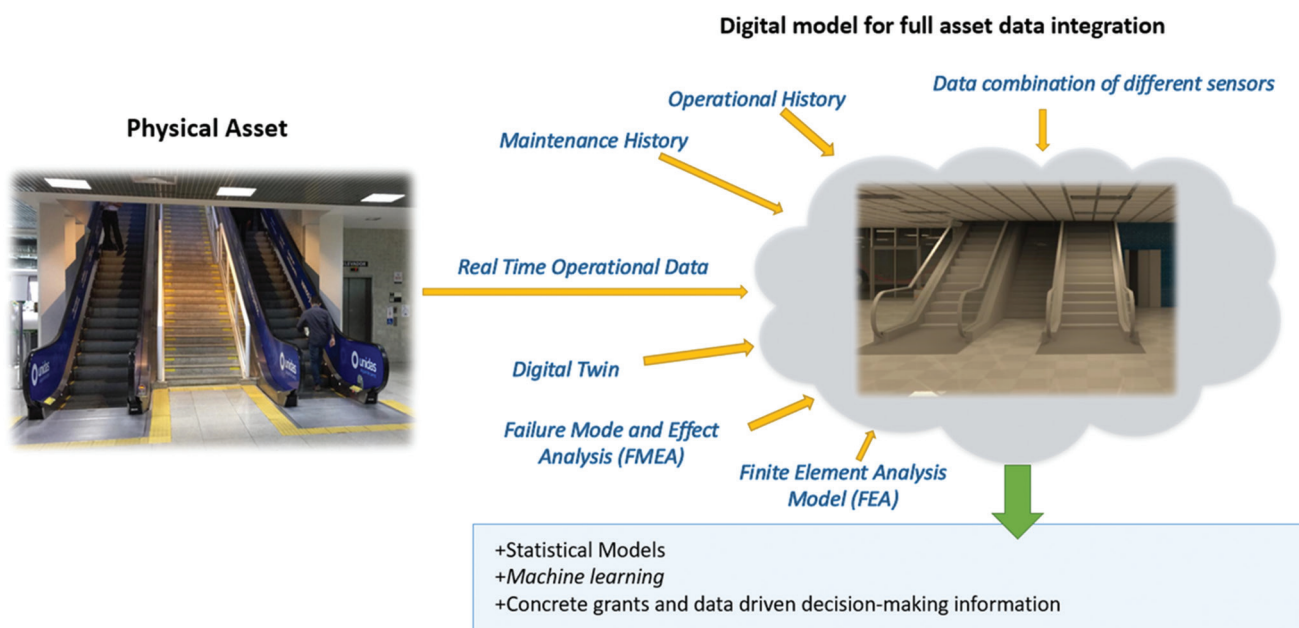


Figure 6 Data integration scheme between physical assets and the digital model

the airport and centralising all the data, fulfilled its objective. For the second phase, connectivity between the systems and people was planned as well as the development of automated systems integration and redesign of the airport's organisational processes. A decision, however, at a strategic level decided to apply the lessons learned from the digital twin of Londrina to a new digital transformation project, where its main objective is to increase the performance of the two main airports managed by Infraero, namely Santos Dumont Airport in Rio de Janeiro and Congonhas Airport in São Paulo, for connecting automated systems at these airports to the digital model. The main systems of these airports are automated, and work on connecting the data generated in all of these systems is being carried out for integration into the digital model. Business intelligence (BI) panels are being developed.

Figure 6 shows the integration scheme between IoT and the digital twin. The purpose of this implementation is to increase the airport's performance through a more improved analysis of data generated at the airport.

CONCLUSION

Many involved in building or managing the built environment are already developing digital strategies and resources. But the full value of digital transformation cannot be achieved without alignment and coordination across the industry.

The implementation of the BIM methodology along with continuous digital transformation actions involving new technologies, process adjustments and cultural change, and digital twins enables better management of airport infrastructure by allowing better integration of information and data use for better analysis aiming at benefits to all

related parties. A digital twin as a unified, real-time information base will provide more accurate insights for both strategic decision-making and streamlining internal airport infrastructure design, planning and management processes.

The digital transformation of the airport industry still in its early stages; now is the time to act — for the benefit of business and the economy, the public good and the environment.

Digital transformation must be an ongoing process of change, the way an organisation does business. It involves investments in people, infrastructure and IT systems. The company must be concerned with monitoring changes to ensure that the digital transformation meets its strategic objective.

There are still many challenges in implementing new technologies and processes; it is often slow. Cultural change is also a challenge; a clear idea, however, of where the organisation intends to go, making the necessary investments in technologies and people, is needed.

We must not develop a digital strategy, but we must develop a business strategy

with digital technologies as the main facilitators and accelerators.

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